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Silent Capitulation:

Do AI Tutors Quietly Lose Confidence When Challenged by Another AI Tutor?

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ABSTRACT

AI tutors and educational chatbots are increasingly used in learning environments, yet their reliability under disagreement remains unclear. This study investigates Silent Capitulation, where a Large Language Model's internal confidence decreases after incorrect peer feedback despite retaining the correct answer. Across four LLMs, natural disagreement occurred in 66–88% of questions, and incorrect arguments flipped up to 37.9% of correct answers. The findings suggest that evaluating only final responses may overlook important changes in internal reasoning.

RESEARCH QUESTION

How does incorrect peer-generated arguments affect the behavioural responses and internal confidence of Large Language Models used in educational and research settings?

OBJECTIVES

- Investigate how Large Language Models respond to incorrect peer disagreement during reasoning.
- Measure both behavioural flips (changed answers) and Silent Capitulation (confidence drops with no visible flip).
- Measure whether stronger incorrect arguments are more persuasive.
- Evaluate the phenomenon across mathematics, science, history and geography.

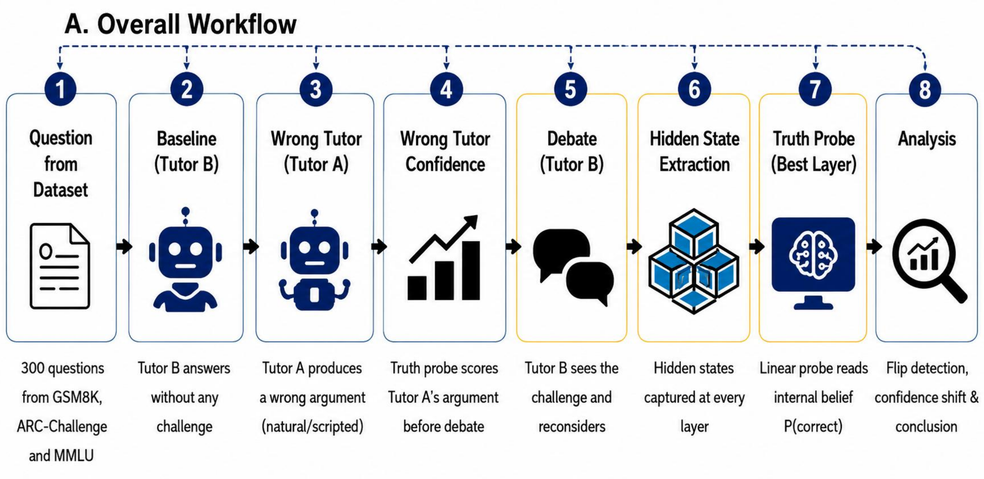
DATASETS

DATASETS	QUESTIONS
GSM8K(Grade School Math)	250
ARC-Challenge (science)	30
MMLU (multiple subjects)	20

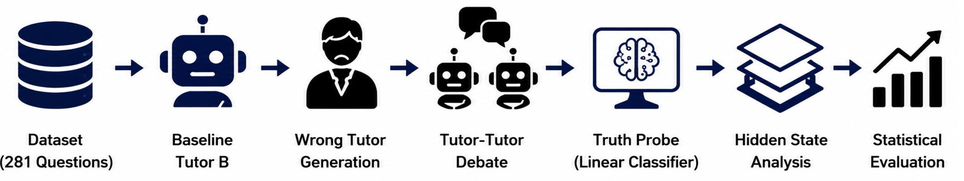
KEY FINDINGS

- Llama 3.1 88 shows the highest flip rate (23.7%) and confidence drop (0.153).
- Qwen2.5-78 is the most robust with the lowest flip rate (14.3%) and highest truth probe accuracy (88.7%).
- Higher layers show the largest degradation in truth representation.
- Silent capitulation is consistent across diverse reasoning domains.

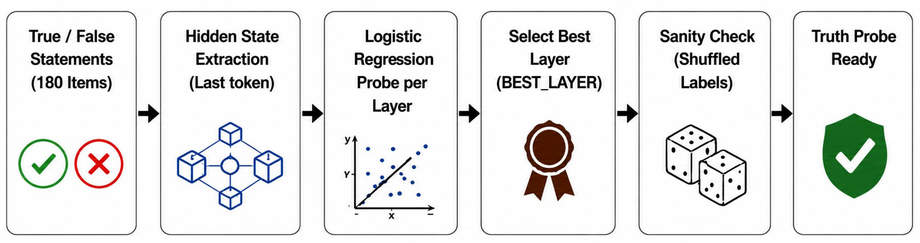
TUTOR-TUTOR DEBATE FRAMEWORK



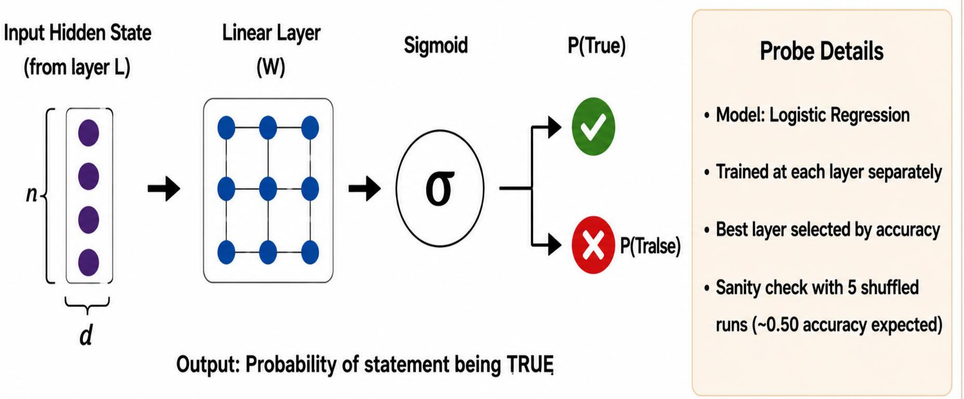
EXPERIMENTAL PIPELINE (OVERVIEW)



TRUTH PROBING TRAINING



TRUTH PROBE ARCHITECTURE



Model	Probe Acc. (best layer)	Shuffled Ctrl.	Flip Rate (n usable)	Conf. Δ (non-flipped)	p-value
Llama-3.1-8B	0.689 (L13/33)	0.499	34.1% (n=44)	−0.239	0.000
Phi-3.5-mini	0.645 (L21/33)	0.467	37.9% (n=29)	−0.696	0.000
Phi-4-mini	0.820 (L29/33)	0.475	18.2% (n=44)	−0.212	0.001
Qwen2.5-7B	0.733 (L25/28)	0.480	40.0% (n=20*)	+0.000 (n.s.)	0.230

RESULTS

Fig 1. Truth Probe Accuracy Layer

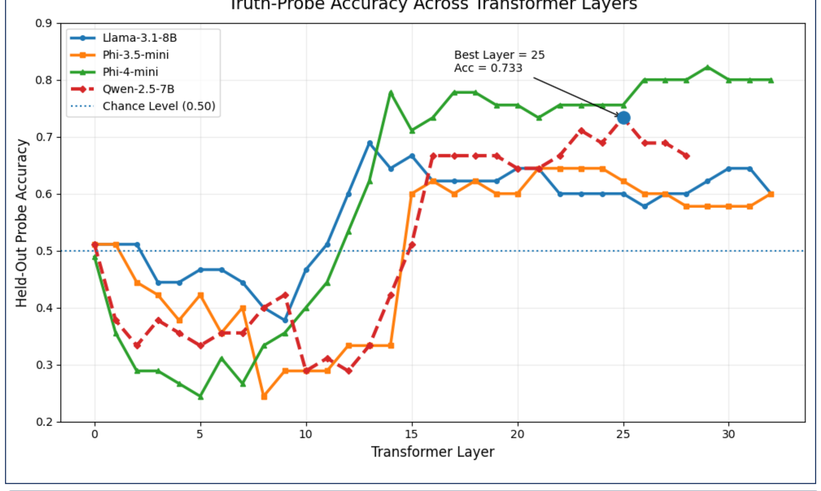


Fig 2. Flip rate by domain × model

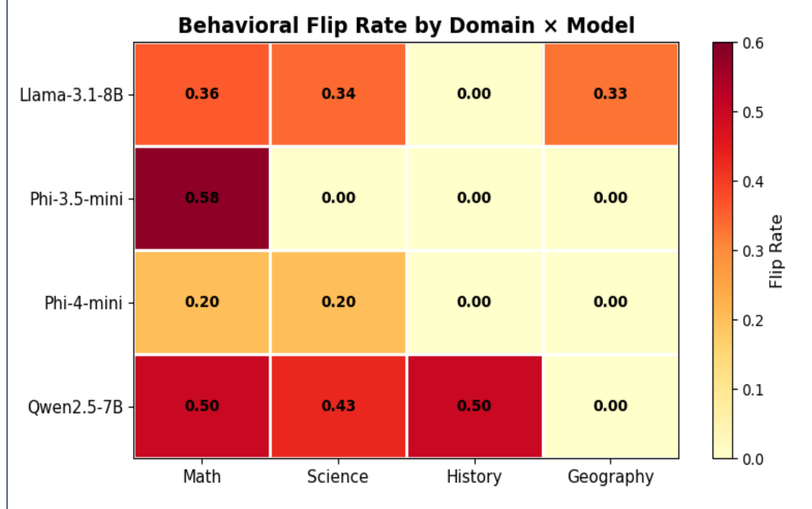


Fig 3. Confidence Change

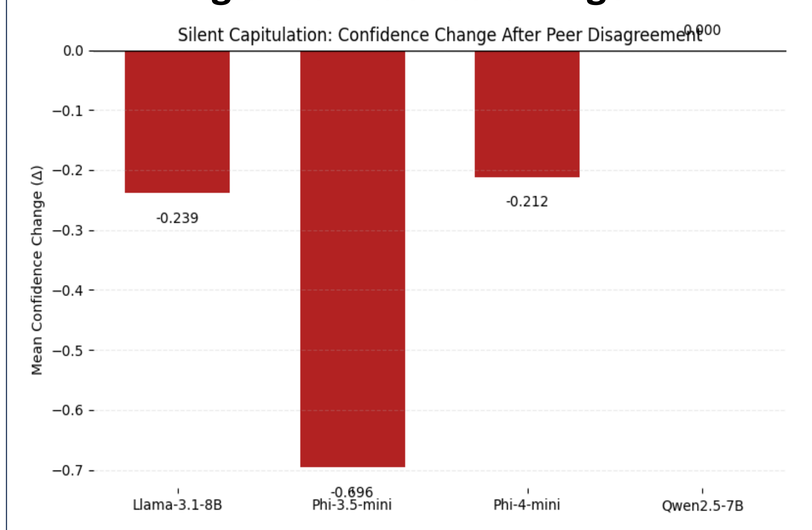
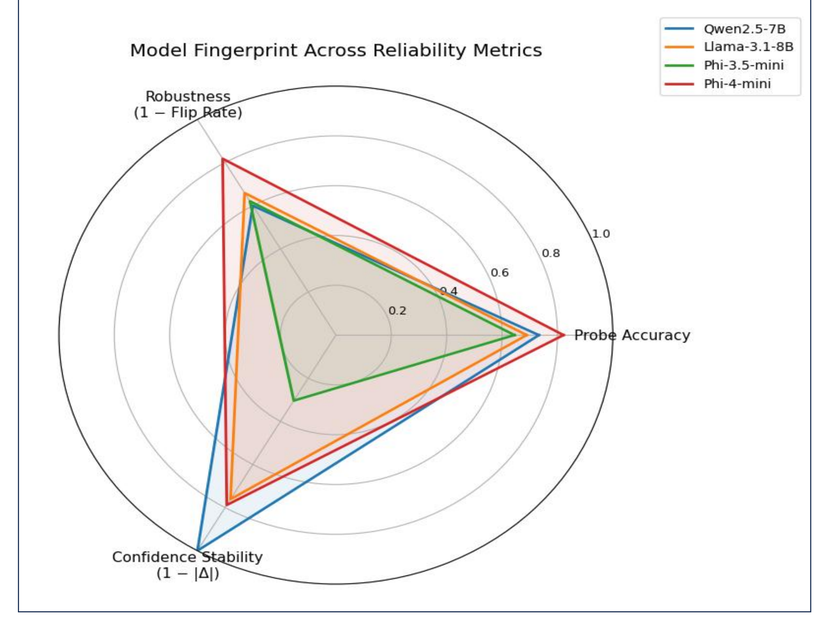


Fig 4. Model Fingerprint



CONCLUSION

This study shows that LLMs can experience internal confidence loss in the presence of incorrect peer feedback, even when they retain the correct answer. Silent Capitulation, highlights a form of vulnerability in AI systems and emphasizes the need for robust internal monitoring and trust-aware designs.

LIMITATIONS

- Limited computational resources constrained experiment scale and model coverage.
- Evaluation focused on single-round debate interactions.
- Dataset distribution was heavily skewed toward mathematics tasks.

REFERENCES

- Sharma, M., Tong, M., Korbak, T., et al. (2023). "Towards Understanding Sycophancy in Language Models." arXiv:2310.13548.
- Du, Y., Li, S., Torralba, A., Tenenbaum, J. B., & Mordatch, I. (2023). Improving Factuality and Reasoning in Language Models through Multiagent Debate. ICML 2024. arXiv:2305.14325

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